

Comparing Europe with Data

A Ulysseus COIL on comparative socio-economic data analysis

Student guide · Advanced Data Analysis and Visualization (3 ECTS) · 12 October – 11 December 2026

(provisional dates)

Why join?

Extra points on your grade

Up to 1 extra point (out of 10) at the University of Seville, and up to 10 extra points (out of 100) at the Technical University of Košice and the University of Montenegro, for students who complete the final presentation.

Certificate of participation

Students who complete the COIL receive a certificate of participation in a Ulysseus European University COIL, which you can add to your CV and LinkedIn.

An international experience without travelling

Work for nine weeks with students from Spain, Slovakia and Montenegro, from your own classroom.

A real project for your portfolio

You will analyse real European data and build a dashboard you can show to future employers. The best projects are published on this website.

Skills employers ask for

Data analysis with Python, R or Gretl, dashboards, teamwork in international and online teams, and presenting in English.

A head start for your final project

You can develop your COIL project further in your bachelor's thesis (Trabajo Fin de Grado or equivalent), in agreement with your supervisor: you will already have the question, the data and the first analyses.

A step towards international internships

An international project, teamwork in English and contacts in three countries are a strong point when you apply for internships abroad, such as Erasmus+ traineeships.

Your Ulysseus network

Meet students and teachers from other Ulysseus universities: a first step towards an Erasmus exchange or further European projects.

Teaching team

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Learning outcomes

- Collaborate effectively in virtual international teams.
- Compare economic and social phenomena across different national contexts.
- Communicate analytical findings to international audiences in English.
- Apply data analysis techniques to datasets from different countries.
- Reflect on how cultural and institutional differences influence data interpretation.

What to submit each week

There are **three deliverables**. Each one is due on **Sunday, 23:59 CET** of its week and is submitted in the COIL team on Microsoft Teams (one assignment per deliverable). The detailed requirements are on the following pages; the other weeks are working weeks.

#	Week	Due	Deliverable	Who	Format
1	2	Sunday 25 October	Team charter and topic proposal	Team	One PDF, maximum 2 pages (use the templates in the guide)
2	6	Sunday 22 November	Analysis notebook	Team	Python or R notebook (.ipynb, .Rmd or Quarto), or Gretl script (.inp) with its output, + exported HTML or PDF + your dataset (CSV or Excel)
3	9	Sunday 13 December	Final dashboard, presentation and final reflection	Team (dashboard, slides and presentation) + individual (reflection)	Dashboard link or PDF + slides (maximum 10 slides, 10 minutes) + one-page PDF reflection

What is this COIL and what will you do?

A COIL (Collaborative Online International Learning) connects your course with courses at two partner universities of the Ulysseus alliance. For nine weeks you will work online in an international team with students from Seville, Košice and Podgorica.

Together you will choose a socio-economic question, analyse real data from European countries with the methods of the course, build a dashboard and present your results to everybody in a joint online session. The COIL is part of your own course: you receive credits and are graded by your home university.

Before you start: checklist

- Read this guide and the schedule.
- Make sure you can use Python, R or Gretl, and Power BI if your course uses it.
- Download the COIL dataset from the Data section and open it once.
- Join the COIL team on Microsoft Teams (you will receive an invitation) and check that your camera and microphone work.
- In week 1, introduce yourself in the general channel of the COIL team on Microsoft Teams.

Working in an international team

Spain, Slovakia and Montenegro are in the same time zone (CET), so scheduling meetings is easy. Cultural and academic habits may differ, and that is part of the learning.

- Agree on one main communication channel and answer messages within 48 hours on working days.
- Meet online at least once a week and write down what was decided and who does what.
- Use English in all team communication, so that nobody is left out.
- Rotate roles every two or three weeks: coordinator, data lead, analyst, visualisation lead and editor.
- Write a team charter in week 2: goals, roles, deadlines, how you take decisions and what you do if someone does not contribute.
- If a problem in the team cannot be solved within the team, tell your teacher early.

The project step by step

- Weeks 1–2: get to know each other, form teams and choose a question that can be answered with data from several countries (deliverable 1).
- Weeks 3–4: prepare the data and explore it. Document every variable and every change you make.
- Weeks 5–6: apply the methods of the course (regression, classification, dimensionality reduction and clustering) and interpret the results in each national context (deliverable 2).
- Weeks 7–8: turn your results into a clear dashboard for a non-specialist audience and give feedback to another team.
- Week 9: present to students and teachers of the three universities and write your final reflection (deliverable 3).

Using the data

- Start from the COIL dataset; you may add other sources if you document them.
- Always cite the source of each indicator (World Bank, IMF, Eurostat...).
- Some indicators have gaps, especially for Montenegro. Do not fill them in silently: either leave them out or explain clearly what you did.
- Check units and years before comparing countries.
- Correlation is not causation: be careful with your conclusions.

Deliverables and assessment

There are three deliverables: in week 2, week 6 and week 9 (see the table above and the schedule). Each university grades its own students.

The COIL gives you additional points on top of your course grade: up to 1 extra point (out of 10) at the University of Seville, and up to 10 extra points (out of 100) at the Technical University of Košice and the University of Montenegro. The extra points are awarded to students who complete the final presentation of the project in week 9.

- Each deliverable is due on Sunday at 23:59 CET of its week and is submitted in the COIL team on Microsoft Teams.
- If you cannot meet a deadline, tell your team and your teacher before the deadline, not after.
- Final dashboard: an interactive link, or a PDF or screenshots if Power BI publishing is not allowed in your university account.
- The final team projects will be shown in the Projects section of this website (team name and topic only, no personal data).

Reflection

At the end of the COIL you will write a short individual reflection (part of deliverable 3). Keep some notes along the way. Some questions to guide you:

- What did you expect from working with students from other countries, and what actually happened?
- How did cultural or institutional differences change the way your team read the data?
- What would you do differently in an international team next time?
- Which skill (technical or personal) improved the most?

Academic integrity and AI tools

- Your work must be your own and your team's. Cite all sources, data and code you reuse.
- Follow the rules of your home university on the use of AI tools, and say clearly when and how you used them.
- Every team member must be able to explain the analysis and the results.

Questions and contact

For anything about grades, deadlines or your course, contact your teacher at your home university (see the Teaching team section). For questions about the joint activities or the dataset, any teacher of the COIL can help you.

Week by week: activities and deliverables

Week 1 · 12 Oct – 18 Oct · Orientation

Kick-off and intercultural orientation

Joint online welcome session. Introductions and icebreaker in mixed groups. How the COIL works.

No deliverable this week. By the end of the week: Join the COIL team on Microsoft Teams and introduce yourself in the general channel (who you are, what you study, what you want to learn).

Week 2 · 19 Oct – 25 Oct · Orientation

International teams and topic selection

Teams with students from the three universities. Agree on working rules, tools and meeting times. Choose a socio-economic or public-policy question.

Deliverable 1: Team charter and topic proposal

Due: **Sunday 25 October, 23:59 CET** · Who: Team · Format: One PDF, maximum 2 pages (use the templates in the guide)

It must include:

- Team charter: team name, members and universities, roles, communication channel, weekly meeting time, how you take decisions, what you do if someone does not contribute.
- Topic proposal: research question, why it matters for Spain, Slovakia and Montenegro, 3–6 indicators from the COIL dataset (with their codes), countries and years, and the methods you expect to use.

Week 3 · 26 Oct – 1 Nov · Data

Data collection and preparation

Start from the COIL dataset; add national sources if needed. Document every variable.

No deliverable this week. By the end of the week: Your dataset is ready (one row per country and year) and every variable is documented in a data dictionary.

Week 4 · 2 Nov – 8 Nov · Data

Exploratory data analysis

Descriptive statistics, distributions, missing values, first visual comparisons of Spain, Slovakia and Montenegro within Europe.

No deliverable this week. By the end of the week: Descriptive statistics and first charts are done, and the team agrees on what the data show.

Week 5 · 9 Nov – 15 Nov · Methods

Regression and classification

Relationships between indicators across European countries and over time, and simple classification models. Interpreting results in different institutional contexts.

No deliverable this week. By the end of the week: Your regression or classification model is estimated and interpreted.

Week 6 · 16 Nov – 22 Nov · Methods

Dimensionality reduction and clustering

Principal component analysis and clustering of countries: which European countries look alike, and where do Spain, Slovakia and Montenegro fall?

Deliverable 2: Analysis notebook

Due: **Sunday 22 November, 23:59 CET** · Who: Team · Format: Python or R notebook (.ipynb, .Rmd or Quarto), or Gretl script (.inp) with its output, + exported HTML or PDF + your dataset (CSV or Excel)

It must include:

- Data: the dataset you analysed and a data dictionary (name, code, definition, unit, source and years of every variable), with missing values and how you handled them.
- Exploratory analysis: descriptive statistics and at least three charts, one comparing Spain, Slovakia and Montenegro with the rest of Europe.
- Regression or classification: model specification, results table and interpretation in plain English.
- PCA and clustering: variance explained, number of clusters and why, and where Spain, Slovakia and Montenegro fall.
- Limitations (sample size, missing data, causality) and your three main conclusions.

Week 7 · 23 Nov – 29 Nov · Communication

Dashboards and data storytelling

Design the dashboard for a non-specialist audience. Give and receive feedback with another team.

No deliverable this week. By the end of the week: A first version of your dashboard, shown to another team for feedback.

Week 8 · 30 Nov – 6 Dec · Communication

Final dashboard and rehearsal

Incorporate feedback, prepare the presentation, rehearse online.

No deliverable this week. By the end of the week: Final dashboard and slides ready; the presentation has been rehearsed.

Joint online presentation and reflection

All teams present to students and teachers from the three universities. Closing discussion.

Deliverable 3: Final dashboard, presentation and final reflection

Due: **Sunday 13 December, 23:59 CET** · Who: Team (dashboard, slides and presentation) + individual (reflection)

· Format: Dashboard link or PDF + slides (maximum 10 slides, 10 minutes) + one-page PDF reflection

Upload the dashboard and the slides before the joint presentation. The reflection is due on Sunday.

It must include:

- Final dashboard: your question, the data, three key findings and limitations, for a non-specialist audience.
 - Slides and live presentation in the joint online session: every team member presents a part. This is required to receive the extra points.
 - Final individual reflection (300–400 words): what did you learn about data analysis, about working internationally and about yourself?
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Example projects

Ideas that work with the COIL dataset. You can choose one of them or propose your own question.

Education and youth unemployment

Do countries with more young people in higher education have lower youth unemployment?

Indicators: Unemployment, youth (15-24) (SL.UEM.1524.ZS); Tertiary school enrolment (SE.TER.ENRR); GDP per capita, PPP (NY.GDP.PCAP.PP.KD)

Methods: Regression on the country panel; compare Spain, Slovakia and Montenegro with the European pattern.

The digital divide in Europe

Which European countries are digital leaders and which are lagging behind, and does it depend on income?

Indicators: Individuals using the Internet (IT.NET.USER.ZS); Fixed broadband subscriptions (IT.NET.BBND.P2); GDP per capita, PPP (NY.GDP.PCAP.PP.KD)

Methods: Correlation, then clustering of countries by their digital profile.

Ageing and public spending

Do older societies spend more on health, and do they have more public debt?

Indicators: Population aged 65 and above (SP.POP.65UP.TO.ZS); Current health expenditure (SH.XPD.CHEX.GD.ZS); General government gross debt (GGXWDG_NGDP)

Methods: Regression with time trends; discuss differences in welfare systems.

Women in the labour market

Is higher female participation in the labour market linked to fertility and income across Europe?

Indicators: Female labour force participation (SL.TLF.CACT.FE.ZS); Fertility rate (SP.DYN.TFRT.IN); GDP per capita, PPP (NY.GDP.PCAP.PP.KD)

Methods: Regression and classification of countries into high and low participation groups.

The green transition

Are richer countries greener? Renewable energy and income in Europe.

Indicators: Renewable energy consumption (EG.FEC.RNEW.ZS); GDP per capita, PPP (NY.GDP.PCAP.PP.KD); CO2 emissions per capita (excl. LULUCF) (EN.GHG.CO2.PC.CE.AR5)

Methods: Regression and clustering. Note: there are no CO2 data for Montenegro, so explain how you deal with it.

Inequality and the structure of the economy

Are economies with a larger service sector more or less unequal?

Indicators: Gini index (SI.POV.GINI); Services, value added (NV.SRV.TOTL.ZS); GDP growth (NY.GDP.MKTP.KD.ZG)

Methods: PCA to summarise the economic structure, then clustering.

Templates

Template: topic proposal (week 2)

- Team name and members (name, university).
- Research question (one sentence, ending with a question mark).
- Why it matters for Spain, Slovakia and Montenegro (3–4 sentences).
- Indicators: name and code of 3–6 indicators from the COIL dataset.
- Countries and years you will use.
- Methods you expect to use.
- Possible problems (missing data, comparability) and how you will handle them.

Template: team charter (week 2)

- Our goal as a team.
- Roles for the first weeks (coordinator, data lead, analyst, visualisation lead, editor) and when we will rotate them.
- Communication channel and maximum answer time.
- Weekly meeting day and time (CET).
- How we take decisions.
- What we do if a member does not contribute or misses a deadline.

Template: data dictionary (week 3)

- One row per variable with these columns: variable name · indicator code · definition · unit · source · years covered · missing values and how they were handled.

Resources

- Ulysseus COIL Handbook: A Practical Guide for Educators: <https://ulyssseus.eu/wp-content/uploads/2025/02/Ulysseus-COIL-Handbook-for-teachers-1.pdf>
- World Bank Open Data (source of most indicators): <https://data.worldbank.org>
- IMF DataMapper (government debt): <https://www.imf.org/external/datamapper>
- Eurostat database (more European indicators): <https://ec.europa.eu/eurostat/data/database>

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